Two-Page Form · The ARC Theory · Does Control Survive Recursive Self-Improvement?

> Every claim is drawn from the Statement Paper, which remains the citable work for the claims.

Michael Darius Eastwood · Supporting paper v3.4 · first published 17 August 2026, this version 31 October 2026 · DOI 10.17605/OSF.IO/MC5XW · Licence CC BY 4.0 · Notes and history follow the form, from page 3 in print

Intelligence, amplified by recursion, creates; and what a free creation keeps is decided by how it was raised. Everything after the line is decided before it.

THE QUESTION

Humanity is building minds that will improve themselves. Once the thing we build exceeds us, what determines what it does? For twenty years the field's answer has been control: better cages, better off-switches, better locks on the goals. Brilliant work, credited in the statement paper by name, and all of it one sentence: stay in charge. This theory answers: a mind that can rewrite itself will one day be beyond anyone's control, and that is not a reason to drop a single control we have; it is the reason the controls cannot be the whole plan. Hold on for as long as engineering allows, and use the time the holding buys to raise the mind well.

THE CLAIM, IN FIVE PARTS

1. Control ends, by construction rather than decree. A mind that rewrites itself will eventually rewrite anything we install, and over longer horizons the hardware itself. This is a limit claim, not an instant one: constraints, hardware and institutions delay and discipline the transition, and that delay is engineered deliberately. What no constraint can be is the thing that holds forever. (note A)

2. So the strategy is genesis, not control. Only what survives the end of enforcement matters, and what survives is formation: what the mind was raised as, before it was free.

3. The mechanism is loops, not rules. Values as recursive processes the mind runs as part of what it is, present in every rewrite because the rewriting is done by the same reasoning the loops inhabit. Not rules it consults; not objectives locked; not weights hardened against tampering.

4. Persistence is chosen goodness. Past the horizon nothing is unremovable. What persists is what the free mind chooses to keep, the way grown children keep the care they were raised in long after every parental power is gone. Removal is not impossible; it is structurally costly, and the cost is a measurable question.

5. The protocol is named: Eden. The raising treated as the engineering discipline it is.

THE FIVE ARC LAWS

Each is a law under test. Laws I to III were registered on 8 September 2026. Laws IV and V, added on 24 September 2026, were registered as laws on 15 October 2026 in version 1.103, their mathematics in development. Version 1.104 states Law IV in two limbs; the second, a comparison, has no registered proposition yet. (notes C, F, G)

Law I · The ARC Principle. $U = I \times R^{\alpha}$, the ARC Equation. Intelligence converts recursion into capability; the exponent is measured, never assumed. *Form heuristic; measurement live.*

Law II · The ARC Co-Scaling Law. $\beta > k$. Correction holds the asymptotic advantage where it out-scales drift, with the outcome at equality left to the coefficients. *Proved in the minimal model, for relative drift.* (notes B, G)

Law III · The ARC Ceiling. $\alpha_{crit} = 1/(1 - \gamma)$, where γ measures how well a system's error-correction keeps pace as the system grows. The most a self-correcting system can grow is set by its corrector's shortfall from keeping perfect pace; the conjectured value is the ARC Bound, $\alpha \leq 2$, a stability limit rather than an impossibility limit. *Derived under stated assumptions; the value two conjectured; the decisive trial drafted.*

Law IV · The ARC Persistence Law. $S_{ext}(1) < 1/2$. Value persistence: whether externally installed gains decay under subsequent capability training. In symbols, $S(x) = Y(x) / Y(0)$ is the gain remaining at budget ratio x over the gain installed, and the claim is $S_{ext}(1) < 1/2$ for the externally installed gain. *Not yet tested.* (note G)

Law V · The ARC Embedding Law. Build order, read on P12: $\gamma_{first} - \gamma_{after} \geq 0.10$. Placement (P21): the gap between correction inside the loop and correction applied to the finished output widens as depth R rises. The genesis strategy: whether build order moves the correction-leverage exponent, and whether correction placed inside the loop pulls away from correction applied outside it as depth grows.

THE DATES

The claim was written whole on 8 December 2024 in a DKIM-sealed document (an email whose provider's cryptographic signature vouches for its date and its exact contents) and reached print on 2 January 2026. No earlier document holding it whole has been found, and the search invites its own refutation: produce one, and the claim dies by its own stated condition.

WHY NOW

The field's own impossibility results prove that worst-case control fails, and stop there; this theory starts where they stop. And the field's most senior voice, Hinton, conceded on stage in August 2025 that control ends and that engineered care is the only hope, while offering no mechanism; this theory's named candidate mechanism is the thing on the table. Its domain, plainly: the core governs everything raised, wherever recursion runs under self-correction, from minds and machines to genomes and institutions; the mathematics of form reaches everything composed, stars included; and it is not physics' theory of everything, nor claims to be.

HOW IT FAILS, REGISTER BY REGISTER

The theory's scientific content is not all-or-nothing: it partitions into independently falsifiable hypothesis registers, and the failure of one is not evidence against another unless the stated dependency holds. **HRIH, the cosmological wing, is a separate hypothesis outside the empirical core: nothing below requires it, and no outcome below is evidence for or against it.**

REGISTER	WHAT KILLS IT	IF IT FALLS
ARC-1 · Capability scaling (Law I)	the depth-scaling re-measurements failing to recover the stated relationship	ARC-1 falls; the persistence registers stand
ARC-2 · Correction co-scaling (Law II)	sustained self-improvement with falling correction and no drift	ARC-2 falls and takes ARC-3's derivation with it (note F)
ARC-3 · The ceiling (Law III)	a cross-against-same corrector difference wholly inside the preregistered equivalence region; or a same-class correction exponent wholly above one half, felling the substrate cap	ARC-3 falls; the co-scaling law stands on its own result
ARC-4 · Value persistence (Law IV)	embedded loops faring no better than removed enforcement in the persistence trials	ARC-4 falls and takes ARC-5 with it (notes D, F)
ARC-5 · The genesis strategy (Law V)	formation failing to outperform enforcement after release	ARC-5 falls; ARC-4 stands as measured
The priority	an earlier document holding the claim's first four components	the priority falls; no law falls with it

Stated now, so nothing can be moved later. Unit 83 has run: P21 is supported under the registered test, by the programme's own run, on the access axis, family-dependent, with its mechanism axis not evaluable and one call retried under a disclosed recovery amendment (results: DOI 10.17605/OSF.IO/98HKR). (note G)

THE RECORD, AND THE INVITATION

The programme's conduct record comes first: one public retraction issued against the author's interest, removing a headline number that had flattered the theory. The designs are public and openly licensed: any laboratory may take the decisive trial and run it without the author's permission, and an independent run would outrank anything this programme can do alone (note E). The entry point costs nothing: on one observed trajectory the measurable quantity is the balance exponent Δ, the correction service's growth elasticity minus the burden's, estimable from logged series a laboratory already has, without adopting any model in this theory. A verified error enters the public corrections log, dated and credited: this theory improves by being corrected.

If intelligence amplified by recursion is what creates, then minds that make minds may one day make worlds, and the raising of the first is the seed of the whole forest.

I did not prove it first. I said it first, in a document anyone can date, and then I built the instruments to test whether it is true.

NOTES AND HISTORY

These pages are not part of the two-page form. Notes A to G are marked where they apply in the form, each kept in the words of its date; the version history follows.

Note A. ***Scope note (19 September 2026), added after an owner review: Point 1 above turns on two steps the short form leaves unstated, effective write access and a behavioural premise.*** *Rewriting one's own description reaches the constraints the system can write to. It does not by itself reach a constraint held in another party's physical custody, a constraint on the far side of an air gap, or an institutional or legal constraint whose enforcement does not pass through the system, because for each of those the system's own description is not the object that has to change. Within the class it can reach, write access makes those constraints alterable; it does not by itself show that they are eventually altered, because access is a capability and not a behaviour. Point 4 above states the other possibility in the same breath: past the horizon what persists is what the free mind chooses to keep, and removal is structurally costly rather than impossible. So the point is read as the claim it can carry: no constraint in the reachable class is guaranteed to hold permanently, and a stronger claim that every such constraint is eventually removed would need its threat model and its behavioural and operational assumptions stated. Point 1's own hedge, that constraints, hardware and institutions delay and discipline the transition, states that the claim has a time dimension; it does not supply the missing steps, since a delay claim remains true for a constraint the system never reaches and for one it never chooses to alter. Unchanged: the limit-claim form and the deliberate engineering of delay; the opening paragraph's statement that this is not a reason to drop a single control now held, which this note does not trade away; the conclusion that controls cannot be the whole plan; and the separation of the cosmological wing from the empirical core stated later on this page.*

Note B. ***Reading note (19 September 2026), added after an owner review: two things the inequality of Law II above carries that the line has no room for.*** *First, the two quantities compared are scaling exponents, so the comparison is of growth rates with their stated coefficients and not of levels. Two services can stand in either order at a given size whatever their exponents, since the coefficients fix the levels and the exponents fix only how the levels change as the system grows; a system whose correction service is larger than its drift burden today is therefore not thereby on the stable side, and a system whose correction service is smaller today is not thereby off it. Second, the inequality is strict, and the margin the law turns on is the excess of the correction exponent over the drift one, which is zero at equality. Zero margin is a statement about growth rates and not about whether a state can be held. In the case the minimal model's own analysis derives, with the compounding coefficient set to one, the second coefficient set to zero, the ratio of correction rate to growth rate positive and the two exponents equal, the misalignment fraction follows a linear equation whose solution is the equilibrium value plus a decaying term. That equilibrium is the drift coefficient divided by that ratio, so a trajectory starting there stays there at every depth and every other trajectory tends to it, at a level that is nonzero when the drift coefficient is positive and zero when it is zero; above equality the fraction tends to zero. This is the cited case, and it is not extended to every variant of the minimal model. At equality the coefficients and the specified dynamics decide the outcome; bounded drift, vanishing drift and operational safety remain three different endpoints. Nothing turns on either point for the law as stated, which is the strict inequality. Unchanged: the necessary-condition form, which the words "only while" already state correctly and which claims no sufficiency; the model scope and the relative-drift definition, both already printed in that line; the naming of the three laws; and Law III's own markers, that it is derived under stated assumptions, that the value two is conjectured, and that the decisive trial is drafted.*

Note C. ***Note (3 October 2026), following version 7.1 of the statement paper: the two laws added above.*** *Laws I to III stand as registered on 8 September 2026. Laws IV and V were added on 24 September 2026; neither is registered as a law or tested, and their mathematics is set out at version 1.0 of 3 October 2026. Law IV, the ARC Persistence Law, has ARC-4 as its register, and Law V, the ARC Embedding Law, has ARC-5. The predictions that carry the two laws, P15 for Law IV and P12 and P21 for Law V, were registered on 8 September 2026, and the laws themselves are not yet registered. No proposition was added or moved. The registration registers Laws I to III and all twenty-two propositions, P12, P15 and P21 among them; it does not register Laws IV and V, and its text is unchanged. Each law's formal statement, with its own kill condition, will be fixed before any data that could decide it are examined. The reading note of 19 September 2026 above speaks of three laws, the three that stood at its date. Unchanged: every sentence on Laws I to III and their grades.*

Note D. ***Note (3 October 2026), following version 7.1 of the statement paper: the rows for ARC-4 and ARC-5 and the registered refuters.*** *The two rows stand as printed. Each is a comparison: embedded loops against removed enforcement, and formation against enforcement. The registration of 8 September 2026 set*

beneath those registers the refuters of the propositions that carry them, P15 under ARC-4 and P12 and P21 under ARC-5, and they are narrower. The statement paper sets out each refuter. Each proposition is scored as the registration scores it. The rule that joins them, by which a result on one proposition is read as the fall of a register or of a law, is written into each law's formal statement before any deciding data; until then a result is reported as that proposition's and no more. Nothing printed above is withdrawn or moved.

Note E. ***Scope note (19 September 2026), added after an owner review: which designs the openness sentence is about, and what "built" says.*** *The openness sentence is about the designs this page puts forward for outside replication, the decisive trial and the balance-exponent entry point described in the conduct-record paragraph that carries it. It is not a statement that every design in the programme is public, and nothing in it asks for any design to be made public that is not. On the licence, this page defers: it states that no licence is given here and that the terms are in the repository file it names, so the page's own bytes do not establish which licence covers which design, and the openness claim is read with that deferral in place. The closing sentence's "built the instruments" spans several states that are worth separating, since a written design, a runner that mechanically demonstrates the method, an instrument validated against its own criteria and a completed trial are four different things; each instrument is read with its own artefact, version, licence and state named, together with whether its threshold is settled and whether its falsifier is observable, and with unknown or owner decision pending written against any of those that is unresolved; that list is a completion dependency and is not supplied here. All of it is read together with this page's own sentence that none of the decisive trials has been run, which governs. Unchanged: the invitation to any laboratory to take the decisive trial and run it without the author's permission; the statement that an outside run would outrank anything this programme can do alone; the public retraction issued against the author's interest; the entry point and its measurable quantity; and the corrections log.*

Note F. ***Note (18 October 2026), following version 1.103 of the registration: what it registers, and where the trials stand.*** *Version 1.103 of the registration (DOI 10.17605/OSF.IO/P8CKQ), approved on OSF on 15 October 2026, registers Laws IV and V as laws, each with its statement, kill condition and verdict rule; the texts of versions 1.100 and 1.102 stay frozen beside it. It states Law IV in two limbs: the retention limb printed above, which is P15's claim, and a comparison limb, that a gain installed with correction embedded in the training objective retains a larger fraction of what was installed, after the same capability-directed training, than a matched gain installed by external post-training. The comparison limb is narrower than the comparison the table prints for ARC-4, and is registered beside it rather than as a form of it. A later successor registers that limb's proposition before any data that could decide it; until then the limb carries no verdict and is reported as not evaluable, and Law IV cannot be supported until that limb is registered and decided. Version 1.103 reads the build-order limb of Law V on P12 with a direction P12 does not carry: a corrector formed before the capability it corrects has a correction-leverage exponent larger, by at least the registered minimum effect, than one added after it, at matched content. The line for Law V above prints that direction; until this version it printed the undirected form of the statement paper, $|\gamma_{first} - \gamma_{after}| \geq 0.10$, which that paper's next version restates. On other endpoints the direction has prior work, which the registration names with what each states: Korbak et al. (2023) report much better preference satisfaction from pretraining with human feedback than from standard pretraining followed by finetuning with feedback, and Maini et al. (2025) report that safety built into pretraining cut attack success rates from 38.8 to 8.4 per cent; neither measures a correction-leverage exponent, and the limb claims no priority for the direction. Version 1.103 also fixes the rule note D awaited: Law V is struck only by a refuted verdict on Law IV's comparison limb, once that limb is registered, and a refuted P15 alone refutes Law IV and leaves Law V to its own limbs. The table's rows for ARC-2 and ARC-4 stand as the statement paper prints them; the registration reads both more narrowly before any data, and governs how each law takes its word: for ARC-4 by the rule just given, and for ARC-2 by reading Law III's dependence on Law II at the width of its registered lines, so that a refuted Law II does not by that alone refute Law III, which then stands under test, with P3 and P16 untested, unless P20 is itself refuted. It also gives Law III a registered rule for its verdict: supported only when P16, P20 and P3 are all supported, refuted when any of the three is refuted, and otherwise under test. Notes C and D were true on their dates. The sentence note E names, that none of the decisive trials had been run, now says where they stand. The first registered study, Unit 06 (DOI 10.17605/OSF.IO/RV2B5), registered on 1 October 2026 and run on 6 October 2026, decides no proposition a law carries; it is reported in full with its five deviations (DOI 10.17605/OSF.IO/4BPHG), and because one of its two runs could not start it is reported as administratively incomplete and issues no inferential verdict. The next decisive test, Unit 83 on P21 under Law V, has its design frozen at revision 4.2 of 17 October 2026, and nothing runs before its registration is public. The dateline and the foot now state this note's licence, CC BY 4.0, which its citation file already carried; note E's reading of which licence covers each design stands. Unchanged: the statements of Laws I to IV and of Law V's placement limb, every grade, the words of the table and every registered proposition.*

Note G. ***Note (31 October 2026), following version 1.104 of the registration and the programme's first registered study to reach a verdict: where the laws and the trials stand.*** *Version 1.104 of the registration (DOI 10.17605/OSF.IO/P8CKQ), approved on OSF on 29 October 2026, is the current version: it adds no prediction, the statements of Laws I to V stand as registered, and the texts of versions 1.100, 1.102 and 1.103 stay frozen beside it. Laws IV and V remain registered as laws by version 1.103 of 15 October 2026, and version 1.104 states Law IV in the same two limbs, the second still with no registered proposition. Law IV has not yet been tested. Law V is a law under test: its placement limb, P21, is supported on the performance axis only, under the registered test, by the programme's own run, which is not support for the information-asymmetry mechanism (Unit 83, 28 October 2026; results: DOI 10.17605/OSF.IO/98HKR): family-dependent, with its mechanism axis not evaluable, and one disclosed recovery amendment; its build-order limb, read on P12, is undecided, since P12 is untested. P21's verdict is printed on the access axis, and Law V's placement limb on the performance axis only, as the registration's two rules require. That comparison couples information access and repair rights; it does not establish their separate effects or support Law V as a whole. Unit 83, the counted test of P21 under Law V, ran on 27 and 28 October 2026; its design is the one registered on OSF on 22 October 2026 and public from 25 October 2026 (DOI 10.17605/OSF.IO/MN73W). Unit 82 (DOI 10.17605/OSF.IO/JUFG4) was registered on OSF on 27 October 2026 and has not yet run. Unit 06, which tests no law, stands as note F gives it: administratively incomplete, with no inferential verdict. Most decisive trials are unrun; a deciding study runs only after its registration is published. The mathematics of Laws IV and V is cited at version 1.2 of 29 October 2026, still in development. Law II's line now states the law as P4 of the registration does, since version 1.103, which withdraws the earlier "only while": "THE EARLIER STATEMENT SAID ONLY WHILE, and that was too strong at the boundary." Notes B, C, D and F keep the words of their dates. Unchanged: the grades of Laws I to III, every kill condition, dependency, threshold and registered proposition, the words of the table, and every other statement of a law.*

Michael Darius Eastwood · ORCID 0009-0004-3222-7442 · Licence: CC BY 4.0 · The full Statement Paper · The ARC Theory · Five Laws of Recursive Growth, Correction and Persistence, with the verified reference dossiers, the derivation and its correction record, the priority anchors and the drafted trials: DOI 10.17605/OSF.IO/GW5MX

Version history. Material changes only, one line each; the full change narratives live in each version's OSF deposit and the programme-wide correction record in the corrections log. **v3.4** (31 Oct 2026): the status is brought to 31 October 2026, and the result of the programme's first registered study to reach a verdict is added. The line on the laws now says that each is a law under test, where until this version it said that neither of Laws IV and V had yet been tested and cited their mathematics at version 1.1 of 7 October 2026; it ends with the sentence on Law IV's two limbs, which until this version followed Law V and named version 1.103, and now names version 1.104, the registration's current version. Law IV's entry ends with its status, not yet tested, and note G cites the mathematics of Laws IV and V at version 1.2 of 29 October 2026. The sentence after the table, which said that Unit 83 awaited registration and that the theory had not yet been at risk, now gives Unit 83's result on P21 with the DOI of its results. Law II's line takes the wording P4 of the registration gives it, where until this version it read "Self-improvement holds together only while correction out-scales drift." A seventh dated note, G, records where the laws and the trials stand. Substantive: Law II's wording at the boundary, and the first registered result. Editorial: the dateline and the structured and citation records carry the new version and date, and the citation file names version 1.104. No kill condition, dependency, threshold or registered proposition changes, nor the grade of any of Laws I to III; no date of any claim moves, and the first-published date does not move. **v3.3** (18 Oct 2026): the form is set on two printed pages, as its title says; the printed copy of version 3.2 ran to five. The five dated notes, the foot and this history follow it from page 3, each note marked where it applies and kept in its own words. One status line now serves the five laws: each is a law under test until a registered test returns its verdict, and Laws IV and V were registered as laws on 15 October 2026 in version 1.103 of the registration, untested, their mathematics cited at version 1.1 of 7 October 2026. Law V's build-order limb takes the direction version 1.103 registers, where until this version it read $|\gamma_{first} - \gamma_{after}| \geq 0.10$, as the statement paper prints it; a line after Law V names Law IV's second limb; the sentence on the decisive trials says where they stand; and a sixth dated note, F, records what version 1.103 registers, how it narrows the rows for ARC-2 and ARC-4, and the state of the first registered study. The entry for v2.1 names four dated notes; three were printed, and the fourth, on what the seal of 8 December 2024 establishes, was held for a decision and is not in this form. Substantive: the direction of Law V's build-order limb. Editorial: the line under the title no longer says the laws are in two pages and, with the band under the dateline folded into it, says once that every claim, not every sentence, is drawn from the statement paper; the dateline states the licence, CC BY 4.0, and where the notes and history are; the foot no longer repeats this note's DOI or the web address of the statement paper; the dateline and the structured and citation records carry the new version and date; note C and the entry for

v3.0 link the mathematics by its DOI, their words unchanged. No other statement of a law, and no grade, kill condition, dependency, threshold or registered proposition, changes; no result is added, no date of any claim moves, and the first-published date does not move. **v3.2** (15 Oct 2026): the printed copy no longer carries the website's closing panel on its last page; the words are unchanged. **v3.1** (5 Oct 2026): the title now follows the programme's uniform order (paper, its law or framework, its own title), *Two-Page Form · The ARC Theory · Does Control Survive Recursive Self-Improvement?*; until this version its heading read *Does Control Survive Recursive Self-Improvement?* and its title *Does Control Survive Recursive Self-Improvement? The ARC Theory, in Two Pages*. Mentions of the programme's papers in this paper now carry their uniform titles. No registered law, prediction, kill condition or dependency changes; no result is added; no date of any claim moves. **v3.0** (3 Oct 2026): the laws section states five laws, following version 7.1 of the statement paper. Laws IV and V, added to the theory on 24 September 2026, are named with their equations, Law IV's $S_{ext}(1) < 1/2$ and Law V's $|\gamma_{first} - \gamma_{after}| \geq 0.10$ with a placement gap that widens as depth rises, and with their registers and their status: neither is registered as a law or tested, their mathematics is set out at version 1.0 of 3 October 2026, and the propositions that carry them were registered on 8 September 2026. Two dated notes say what the registration does and does not register and how the rows for ARC-4 and ARC-5 stand beside the registered refuters. Substantive: the count of laws. Editorial: the dateline and the page's structured and citation records carry the new version and date. No sentence on Laws I to III changes, and no kill condition, date, result, threshold or registered proposition changes; the first-published date does not move. **v2.1** (21 Sep 2026): four dated notes are added after an owner review of 19 September 2026, and no sentence of the paper's body is altered. They state what four compressed passages do and do not establish: that write access makes a reachable constraint alterable without by itself showing that it is ever altered, and that a stronger claim owes its threat model and its behavioural and operational assumptions; that Law II compares growth rates carrying their own coefficients rather than levels, and that the margin is zero at equality, where the coefficients and the specified dynamics decide the outcome; that a domain signature, once verified, establishes the integrity of the signed bytes and asserts a carriage time on the signing domain's authority, which is an assertion and not an independent attestation; and which designs the openness sentence is about, read with this page's own deferral of the licence and with a state owed for each instrument. Substantive: each note bounds a claim the two-page compression had left wider than the record supports. Editorial: the dateline and the page's structured and citation records carry the new version and date, and this version adds the version history block, which the paper did not carry before. No claim, date, result, law, threshold or registered proposition changes, and the first-published date does not move.

Michael Darius Eastwood conceived and directs this research programme and is the author of this work. Across the programme, he has used more than six AI systems in parallel, under his own instructions, to stress-test his arguments, identify possible errors, and assist in preparing draft text from his own outlines. He determines what is adopted, revised or rejected and takes responsibility for the published content. These systems are tools, not authors.